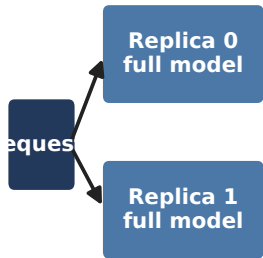


Data Parallelism (DP)

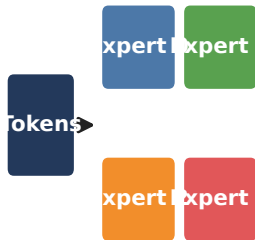


Tensor Parallelism (TP)



Collectives across shards

Expert Parallelism (EP)



Top-k routing + all-to-all